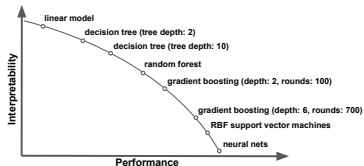
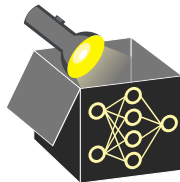


Interpretable Machine Learning

Intro to IML

Introduction, Motivation, and History

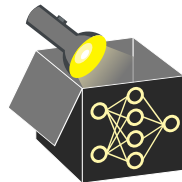


Learning goals

- Why interpretability?
- Developments until now?
- Use cases for interpretability

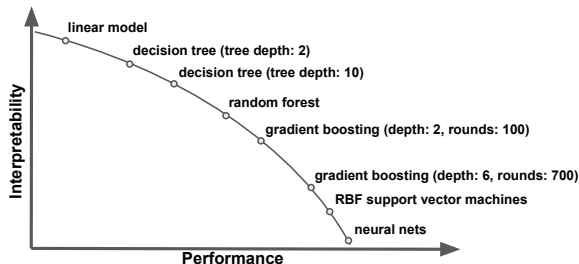
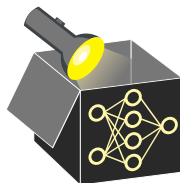
WHY INTERPRETABILITY?

- ML aids decision-making in applications through strong predictive performance
- ML models are black boxes, e.g., XGBoost, RBF SVM, or DNNs
 $\rightsquigarrow$ too complex to be understood by humans
- Some applications are “learn to understand”, not “learn to predict”



WHY INTERPRETABILITY?

- ML aids decision-making in applications through strong predictive performance
 - ML models are black boxes, e.g., XGBoost, RBF SVM, or DNNs
→ too complex to be understood by humans
 - Some applications are “learn to understand”, not “learn to predict”
 - When deploying ML models, lack of explanations
 - 1 reduces trust
 - 2 creates barriers to adoption
- Many disciplines with required trust rely on traditional models, e.g., linear models, despite lower predictive performance



INTERPRETABILITY IN HIGH-STAKES DECISIONS

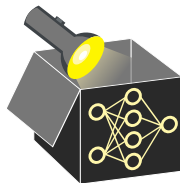
Examples of critical areas where ML-based decisions can affect human life

- Credit scoring and insurance

applications

► "Society of Actuaries" 2021

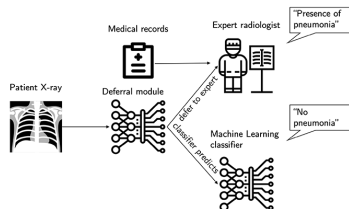
- Reasons for not granting a loan
- Fraud detection in insurance claims



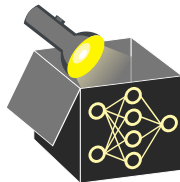
INTERPRETABILITY IN HIGH-STAKES DECISIONS

Examples of critical areas where ML-based decisions can affect human life

- Credit scoring and insurance applications ▶ "Society of Actuaries" 2021
 - Reasons for not granting a loan
 - Fraud detection in insurance claims
- Medical applications
 - Identification of diseases
 - Recommendations of treatments
- ...



▶ [Click for source](#)



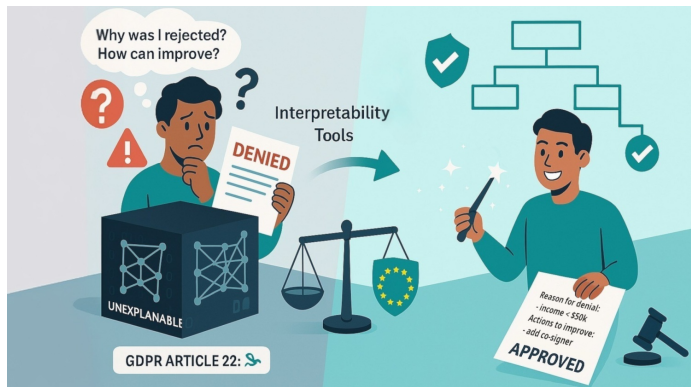
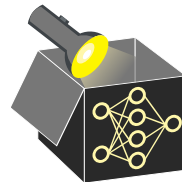
NEED FOR INTERPRETABILITY

Need for interpretability becoming increasingly important from a legal perspective

- General Data Protection Regulation (GDPR) requires for some applications that models have to be explainable ▶ “Goodman and Flaxman” 2017

~> *EU Regulations on Algorithmic Decision-Making and a “Right to Explanation”*

- *Ethics guidelines for trustworthy AI* ▶ “European Commission” 2019



BRIEF HISTORY OF INTERPRETABILITY

- 18th/19th century:
Linear models (Gauss, Legendre, Quetelet)
- Mid-20th century:
Emergence of sensitivity analysis
- Mid- to late 20th century:
Rule-based ML, incl. decision rules and trees
- Early 2000s:
Interpretation tools for ensembles:
 - variable importance for random forests
 - partial dependence for gradient boosting
- Early 2010s:
Explainability methods for deep learning
- Late 2010s:
IML / XAI as a distinct research area



► "Carl Friedrich Gauss" 1828

► "Wikipedia" 1777

